



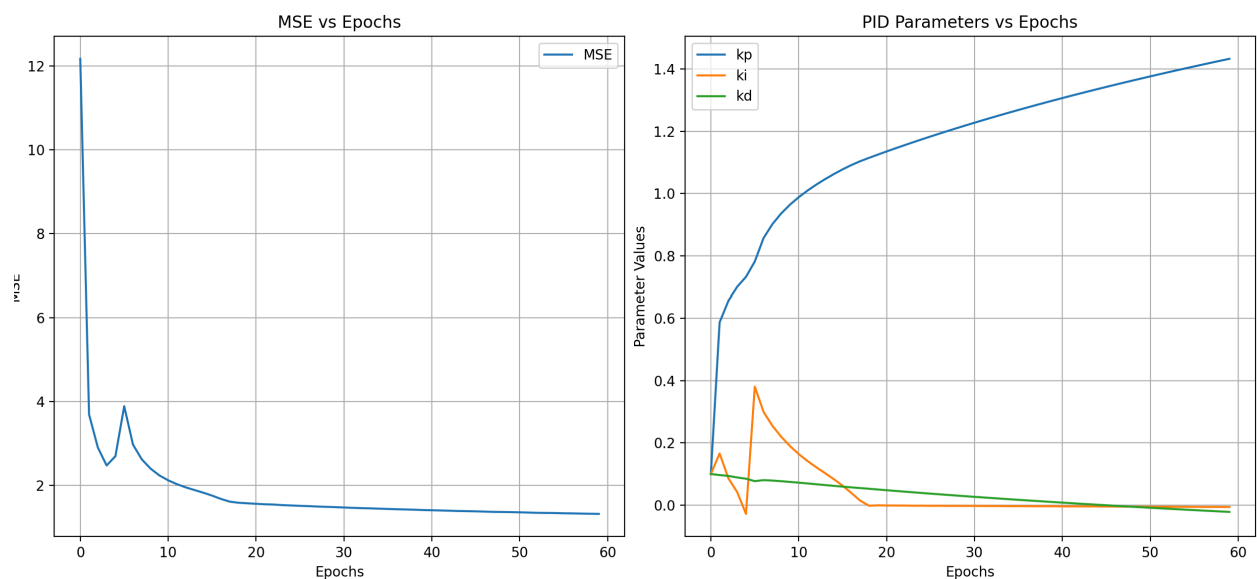
JAX rapport

Bathtub model

Classic PID controller

In this run we can see the k_p parameter quickly increasing marking its significance in the loss of the run. The other two parameters maintain somewhat the same value except for k_i when it spikes. This also is shown in the spike of the loss-function. This might be to the water height being set quite high, so the controller waits until it drains before filling it again.

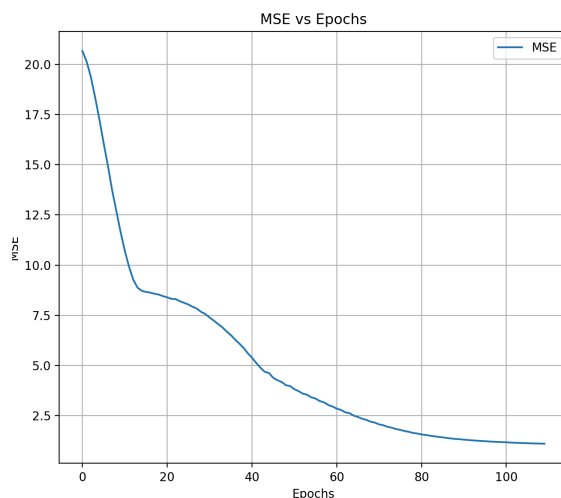
Controller	Classic PID	$K_i = 0.1, K_p = 0.1, K_d = 0.1$
Epochs	60	
Timesteps	25	
Lr	0.01	
Plant	Bathtub	$\text{area_tub} = 3.0, \text{area_drain} = 0.01, \text{water_hieght} = 15.0, \text{target} = 10.0$



NN controller

The NN starts off quite good by having a steep learning before it slowly decreases. Here the target is also like in the PID-controller a bit of the initial height which could make it harder for the NN to learn. Needed 100+ epochs to see the curve flatten.

Controller	NN Controller	activation = [relu, relu], layers = [4,4], weight_range = [-1.0,1.0], bias_range = [0.0,1.0]
Epochs	110	
Timesteps	25	
Lr	0.01	
Plant	Bathtub	area_tub = 3.0, area_drain = 0.01, water_hieght = 15.0, target = 10.0

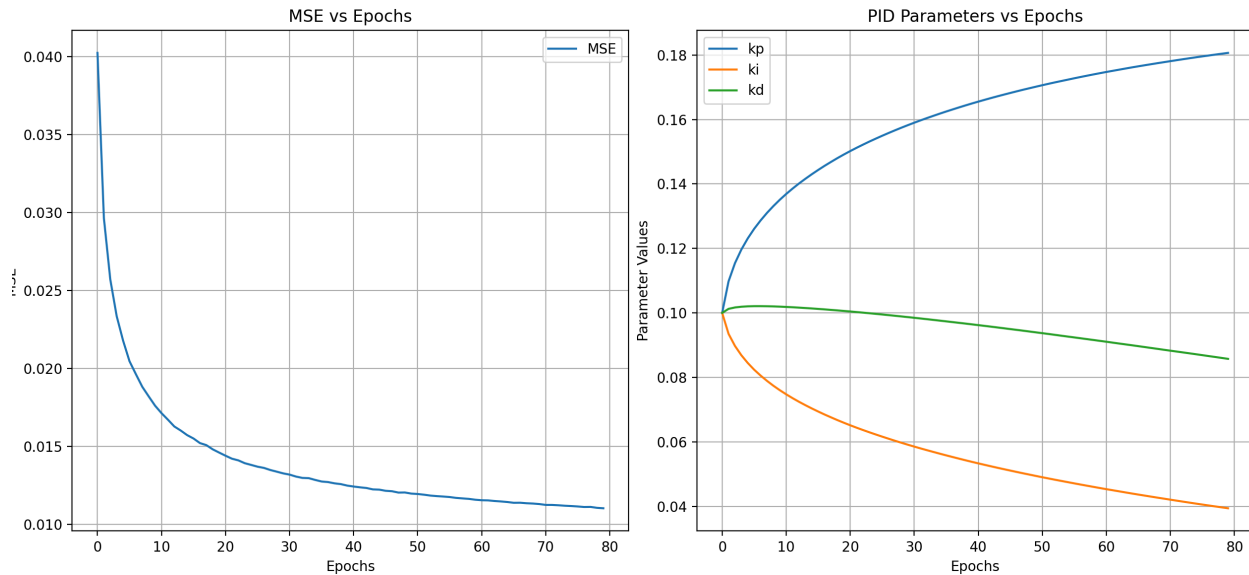


Cournot model

Classic PID Controller

This run shows the PID controller learning very fast and setting its parametres to their assigned values almost straight away. It is obvious that k_p and k_i were the parametres most far off the initial value.

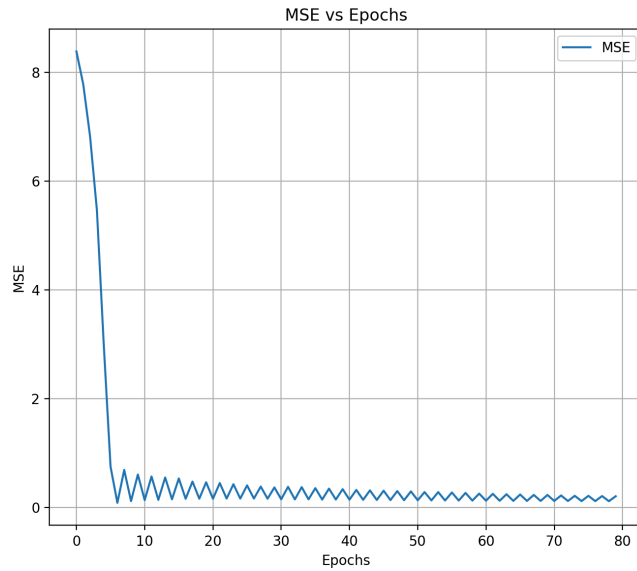
Controller	Classic PID	Ki = 0.1, Kp = 0.1, Kd = 0.1
Epochs	80	
Timesteps	25	
Lr	0.01	
Plant	Cournot	pmax = 5.0, marginal_cost = 0.1, target = 0.5



NN Controller

This run shows very fast learning by the NN but when the MSE reaches a certain level the saw-tooth pattern appears the model's loss is oscillating rather than steadily converging. This is an indication that the learning rate is too low / high at the end, and we could here have implemented a cyclic or step-decay learning rate scheduler.

Controller	NN Controller	activation = [relu, relu, relu], layers = [12,8,8], weight_range = [-1.0,1.0], bias_range = [0.0,1.0]
Epochs	80	
Timesteps	25	
Lr	0.01	
Plant	Cournot	pmax = 5.0, marginal_cost = 0.1, target = 0.5



Fuel tank model

This plant models represents a **fuel tank system**, where fuel consumption and disturbances affect the fuel level over time.

u_t : external refueling

d_t : random environmental effects

F_t : fuel level

k : fuel consumption rate

Updates as following:

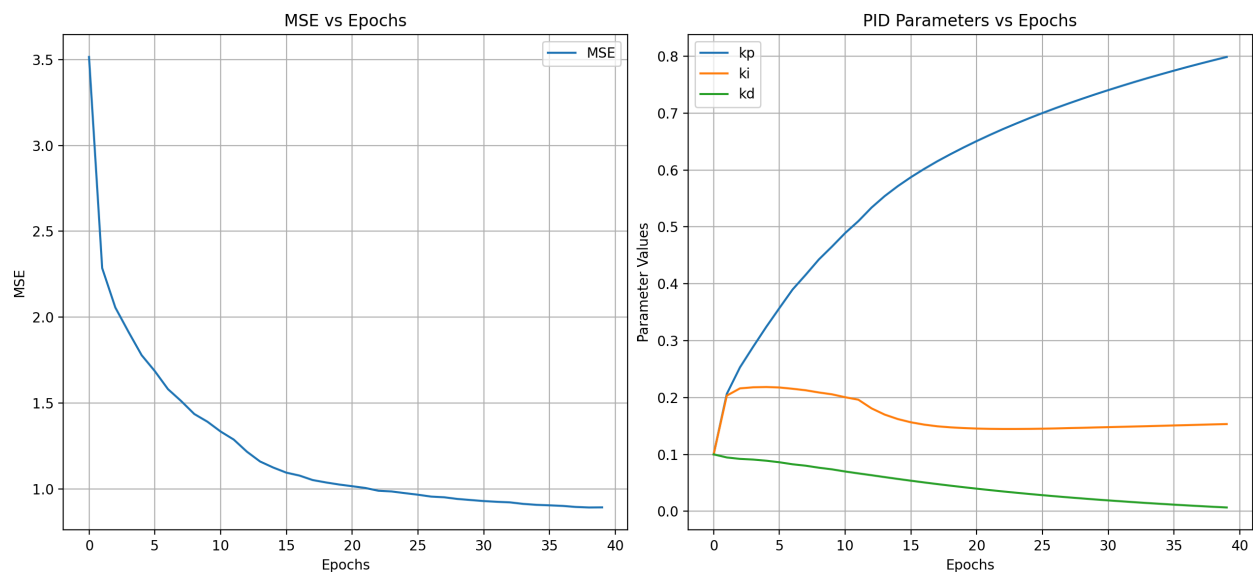
$$\frac{dF}{dt} = F_t + u_t + d_t - k * F_t$$

Classic PID Controller

This run shows quick learning from the controller. The k_p parameter quickly increase its value, while the k_i and k_d parametre mainly keeps their values close to the initial value.

Controller	Classic PID	$K_i = 0.1, K_p = 0.1, K_d = 0.1$
Epochs	40	
Timesteps	25	
Lr	0.01	

Plant	Fueltank	consumption_rate = 0.1, initial_fuel = 5.0, target = 9.0, max_cap = 9.0
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NN Controller

In this run I initialized the NN Controller with four layers and three different activation functions. The loss quickly falls, and the NN learns quite quickly.

Controller	NN Controller	activation = [relu, relu, tanh, sigmoid], layers = [16,4,6,6], weight_range = [-1.0,1.0], bias_range = [0.0,1.0]
Epochs	70	
Timesteps	25	
Lr	0.01	
Plant	Fueltank	consumption_rate = 0.1, initial_fuel = 5.0, target = 9.0, max_cap = 9.0

